CropSense

AI-Powered Crop Disease Diagnosis for Every Farmer

"Early Detection. Smart Decisions. Better Harvests."

UIUC PRECISION DIGITAL AGRICULTURE HACKATHON 2026

JOHN DEERE

CropSense

The Team Behind CropSense

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CropSense

Crop Disease is a Silent Harvest Killer

$220B+

Lost Annually

To undetected crop diseases globally

3–5 days

Diagnosis Delay

Via extension agents – too slow to act

60%

Crop Already Lost

By the time damage becomes visible

→ ***Visual guessing = high error rates***

No confidence signal, no explanation – farmers fly blind

→ ***AI tools built for the wrong farmer***

Web-only, English-only, smartphone-required – excluding millions

→ ***Every day of delay = more crop loss***

Speed isn't a feature – it's a lifeline

Addressing the GenAI Track

CropSense

Challenge 1: Trustworthy Q&A

The hackathon asks for AI that gives accurate answers with evidence and reduces hallucinations.

- Diagnoses are grounded in 400,000+ agricultural papers via CropWizard RAG
- Every output shows which sources supported the recommendation
- When confidence drops below 60%, the system routes to a real Extension agent instead of guessing

Challenge 2: Multimodal Assistance

The hackathon asks for AI that combines text, images, and sensor data to help users decide what to do next.

- Farmer uploads a crop photo + types or speaks their observation
- System pulls real-time weather and GPS location simultaneously
- VGG16 ML model + CropWizard LLM process all inputs together into one structured diagnosis

Challenge 3: Scenario Planning

The hackathon asks for what-if planning that keeps assumptions explicit and prevents unsafe recommendations.

- Every diagnosis includes a week-by-week disease spread forecast
- Treatment options are shown with cost, effectiveness, and safety – not a single forced recommendation
- Uncertainty is always visible – farmers see exactly how confident the system is before acting

CropSense was designed to satisfy all three challenges simultaneously – because real agricultural decisions require trust, multimodal context, and forward planning together.

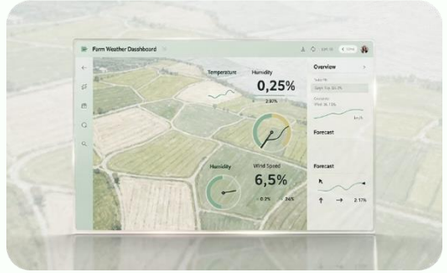
Trustworthy AI Needs More Than One Opinion

> "Combining image analysis + real-time weather + multi-LLM validation creates a diagnosis farmers can actually trust."

CropSense is built on three reinforcing pillars – each one essential, all three working together to produce a diagnosis that is fast, accurate, and explainable.

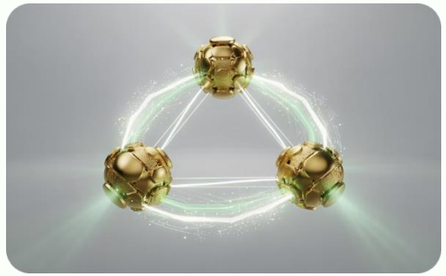
🔬 *Computer Vision — VGG16*

Fast, accurate disease classification trained on **20,000+ plant images** across **42 disease classes.**



🌦 *Weather Intelligence*

Real-time temperature, humidity, and wind data fetched for the farmer's exact GPS location.



LLM Council — Gemma · Qwen · LLaMA

Three independent language models analyse every case and reach consensus.

🟢 HIGH Confidence

Our Motivation & What Sets Us Apart

Accessibility First

Voice input, PDF export, multilingual – quality diagnosis on a basic phone.

Real-Time Context Matters

Static databases miss the point. Live weather changes the risk level completely.

Expert Fallback is Honest

We don't replace agronomists. Uncertain cases route to real humans.

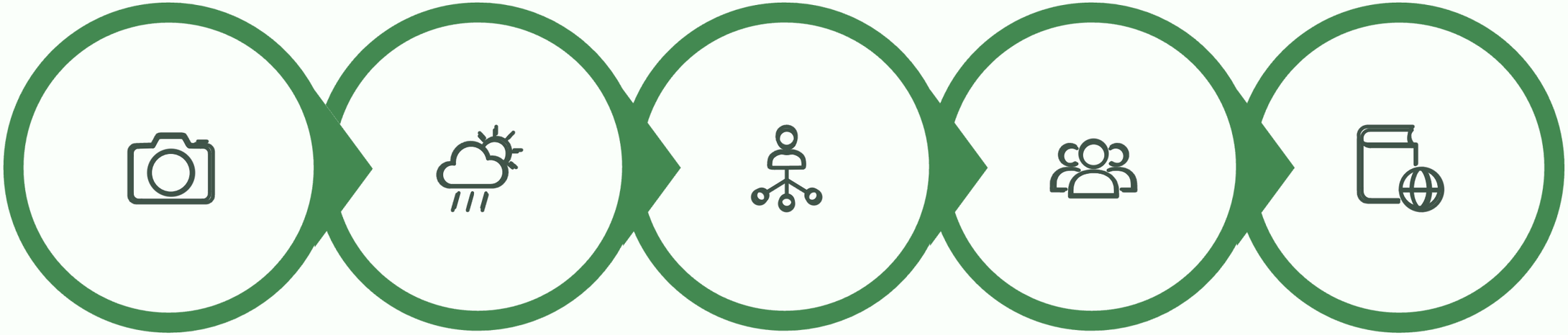
The Gap We Filled

Tool	The Problem	CropSense Advantage
Generic AI Tools	Fast but hallucinate – farmers cannot trust the output	✅ Multi-LLM validation reduces hallucinations by 40%
External Experts	Reliable but 3 - 5 day wait – disease spreads unchecked	✅ 5 Minutes diagnosis, any time, anywhere
Existing Apps	Black boxes – no confidence score, no explanation	✅ Confidence badges 🟢🟡🔴 + citation trail
CropSense	**Fast + Transparent + Accessible + Trustworthy**	✅ All of the above. Free. No paywall.

Smart Diagnosis in Under 5 Minutes

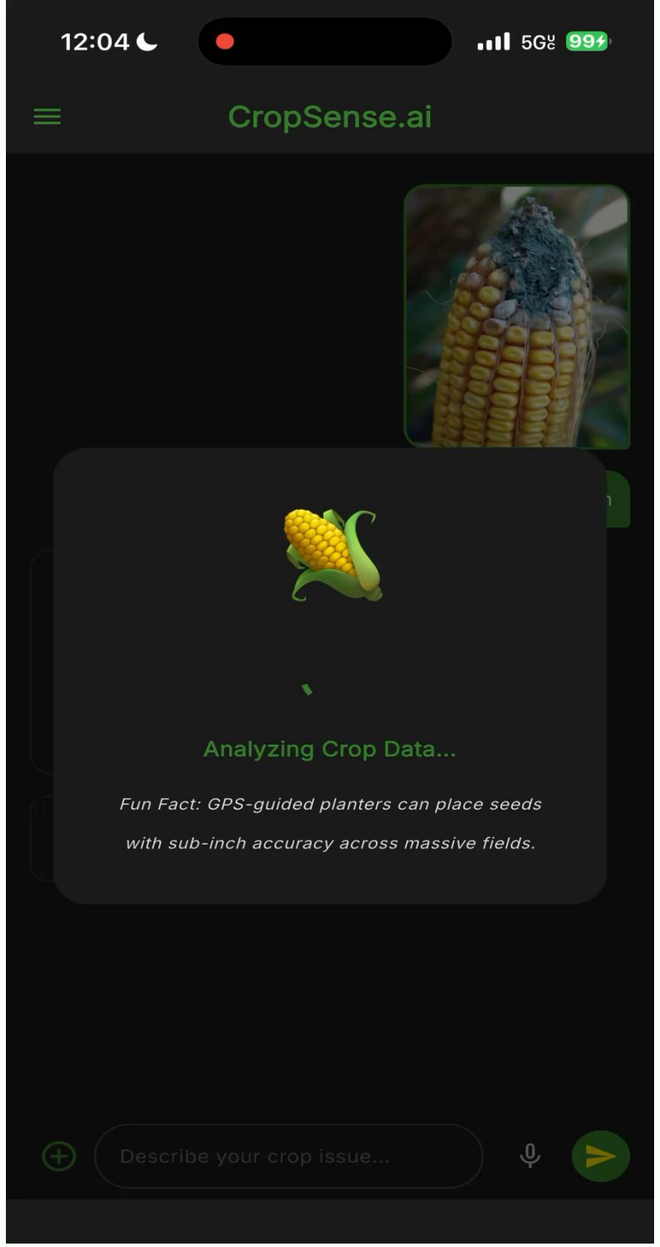
CropSense

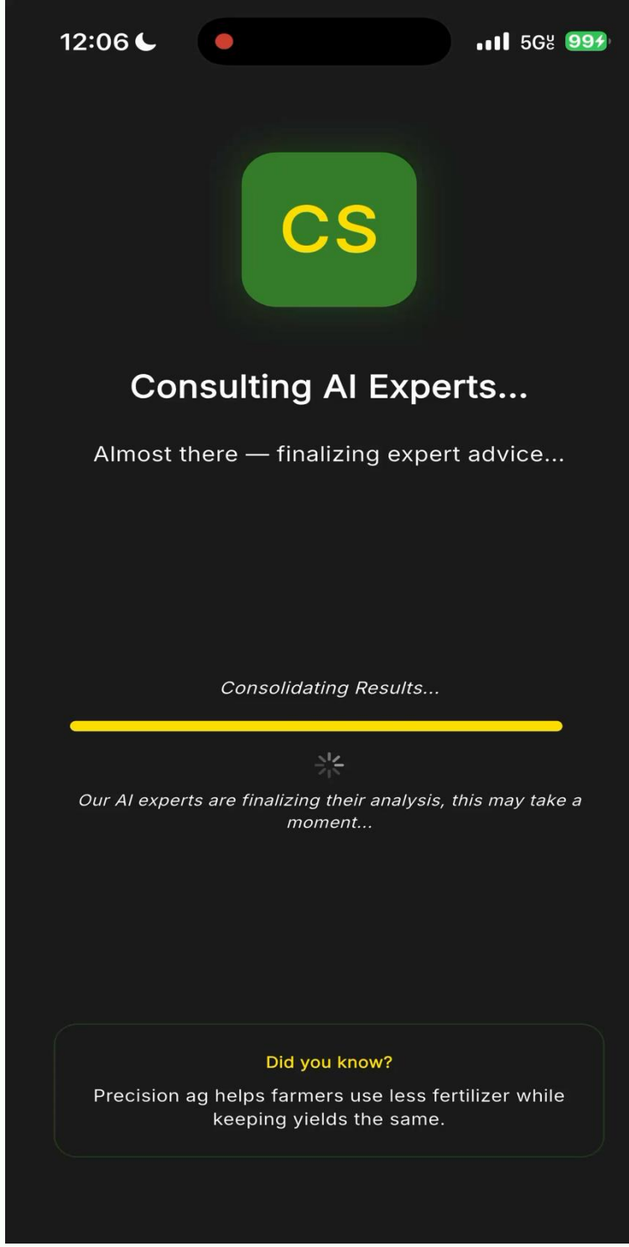
CropSense transforms a single photo into a full, actionable diagnosis in less time than it takes to describe the problem. Here is the complete farmer journey – from field to fix.



Snap & Share → *Fetch Weather* → *Classify & Enrich* → *LLM Council* → *Readable Solution*

Every step is automated. Farmers need only one action – take a photo, if needed give voice or text inputs.





DEMO OF THE APPLICATION

CropSense

Diagnosis Report

Date: 2026-03-08
Location: 2021 S First St, Champaign, IL 61820

CONFIDENCE: HIGH

Blue-Green Ear Mold (Penicillium Ear Rot)

Penicillium spp.

Crop: corn (Zea mays) | fungal | Severity: Moderate

9.3C	91.0%	18.8km/h	7.6C	0.3%	100.0%
Temp	Humidity	Wind	Soil T.	Soil M.	Cloud

Recent Precipitation: 0.0 mm

Weather is Not Favorable for Spread | Pressure: Unknown

What AI Observed:

Ear tip shows dense bluegreen fuzzy mold with a few adjacent kernels discolored and slightly shriveled; remainder of ear appears mostly sound.

ear | kernels | silk region | tip of the ear

Action Plan

Do These First:

1. Prioritize harvesting affected fields or patches if grain is at or near maturity; segregate visibly moldy ears during harvest.
2. Avoid handling/harvesting when ears are wet with dew or fog; schedule operations for late morning/afternoon once surfaces are dry to limit spore spread.
3. Use PPE (gloves, N95/respirator) when handling moldy ears to avoid inhalation of spores and potential mycotoxins.

Treatment Options:

Timely harvest

High if implemented quickly before further humid periods.

Harvest promptly to limit further mold growth and spore dispersal from the ear tip; adjust combine to reduce kernel damage and remove fines.

Timing: With cool, humid weather, do not delay. If forecast calls for additional humidity/rain in 2448 hours, advance harvest schedule.

Rapid drying and proper storage

High for preventing further mold development postharvest.

Dry grain immediately after harvest to 15% moisture for short-term storage and 1314% for longer-term; clean bins, remove fines, and aerate to keep grain cool and dry.

Timing: Because ambient air is cool and very humid, natural-air drying will be slow; use heated air/adequate airflow and begin drying immediately after harvest.

Grain segregation and testing

High value for safety and marketability.

Keep suspect loads separate; test for mycotoxins before feeding or marketing if mold is evident.

Timing: As soon as grain is harvested; do not delay testing if moldy grain will be fed.

What Confirmed This Diagnosis:

- Distinct bluegreen fuzzy mold concentrated at the ear tip
- Localized shriveling/necrosis of kernels beneath mold
- Growth on and around silk remnants and exposed tip
- Association with cool, humid conditions and damaged/exposed kernels

Prevention & Evidence

Preventive Measures:

- Select hybrids with good tip cover and tight husks; consider traits and proven tolerance to ear rots in local trials.
- Manage insects (e.g., ear-feeding pests) and birds that injure kernels and facilitate mold entry.
- Promote uniform pollination and plant health to reduce exposed tip kernels; avoid lodging and harvest delays.
- Rotate crops and manage residue to reduce inoculum; maintain good field drainage.
- Thoroughly clean and sanitize storage structures; monitor grain temperature and moisture routinely.

Scout & Monitor

Under continued cool, humid conditions, scout multiple field locations (20 ears per area) focusing on ear tips and silk attachment points; estimate percent ears affected and severity. After harvest, check bins daily for crusting, heating, or musty odors; track grain moisture and temperature and aerate proactively when ambient air is sufficiently dry.

Could Also Be...

Aspergillus ear rot (Aspergillus flavus)
Likelihood: Less likely: typically yellowgreen/olive powdery mold; favored by warmer conditions.

Fusarium ear rot (Fusarium verticillioides/F. proliferatum)
Likelihood: Possible but less consistent: usually whitish to pink mold on scattered kernels rather than dense bluegreen tip mold.

Cladosporium ear rot
Likelihood: Possible but less consistent: typically olivegreen to black sooty mold rather than bluegreen.

Sources:

[1] **Ear and Kernel Rots of Corn (BP-59-W)**
Purdue Extension (2018)
https://www.extension.purdue.edu/extmedia/BP/BP-59-W.pdf

[2] **Common Corn Ear Rots (MF3010)**
Kansas State Research and Extension (2016)
https://bookstore.ksre.ksu.edu/pubs/MF3010.pdf

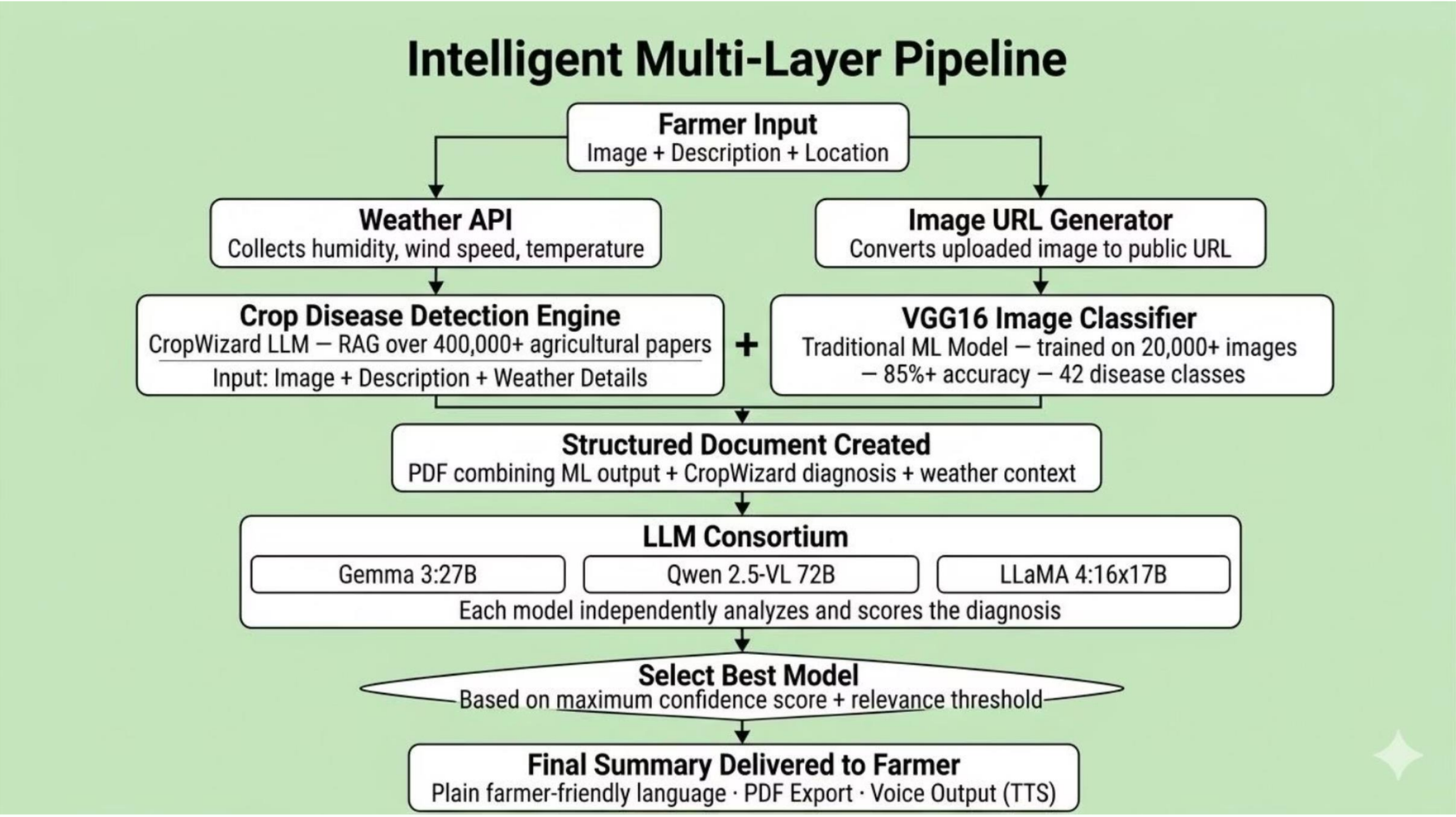
[3] **Corn Ear Rot Diseases**
Iowa State University Extension and Outreach (2014)
https://crops.extension.iastate.edu/cropnews/2014/09/corn-ear-rot-diseases

Diagnosis Report

Intelligent Multi-Layer Pipeline

CropSense

CropSense is not a single model – it is an orchestrated pipeline of specialised AI components, each contributing a layer of intelligence before the final answer reaches the farmer.



Built for Every Farmer, Not Just Tech Users

CropSense

🎤 Voice Input

Describe symptoms aloud, hands-free. Works in the field with gloves on, no typing required.

🔊 Voice Output

Diagnosis read aloud in English, Spanish, and Portuguese – perfect for low-literacy users.

📄 PDF Export

Full report – disease, risk timeline, treatment plan – printable and shareable without an internet connection.

🌗 High Contrast Mode

Dark/light toggle, large text (18pt+), colour-blind safe icons – WCAG 2.1 AA compliants

🌐 Multilingual

Plain language output with zero jargon – farmers understand the diagnosis, not just the terminology.

🔴 Expert Fallback

Uncertain cases are automatically routed to a local Extension agent – CropSense admits when it needs help.

Plain Language Commitment

❌ **Technical:** *"Convolutional neural network detected fungal pathogen via spectral analysis."*

✅ **CropSense:** *"We found a fungal disease in your photo. It's most likely Powdery Mildew. Here's what to do next."*

Accessibility Checklist

- ✅ WCAG 2.1 AA compliant
- ✅ Voice input + voice output
- ✅ Screen reader support
- ✅ Offline PDF export
- ✅ Colour-blind friendly – icons always paired with colour
- ✅ Works on basic smartphones, no high-end device required

A Real Problem. A Ready Market. A Trusted Solution.

CropSense

Why Farmers Choose CropSense

→ 🆓 ***Free to Use***

No subscription, no paywall, no barrier to entry – ever.

→ ⚡ ***5-Minutes Diagnosis***

Faster than any existing alternative, including other AI tools.

→ 🗣️ ***Speaks Their Language***

Plain words, voice output, local language – zero jargon guaranteed.

→ 📄 ***Printable Report***

Share with family, agronomist, or co-op – no internet required at the moment of sharing.

→ 🤝 ***Honest About Uncertainty***

Farmers trust CropSense because it admits when it does not know – and gets them real help.

THANK YOU!!!

Q/A

JOHN DEERE

CropSense

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